Combinatorial geometry

L5-X7 Level 5 Advanced · Combinatorics · Beyond the textbook

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Extremal choices and Sylvester-Gallai, Erdős-Szekeres, rotating lines and ham sandwich cuts, Euler's formula and the crossing lemma, Helly's theorem, and area arguments with lattice points, at BMO2 and IMO level.
By the end you can make extremal choices of points, lines and triangles, and prove Sylvester-Gallai and Erdős-Szekeres type results with them; run rotating-line and discrete continuity arguments, and bound segments and crossings with Euler's formula and the crossing lemma; find the convex family for Helly's theorem and use area averaging with lattice points.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1), Graph theory and extremal combinatorics (L5-CO1). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Combinatorial geometry asks what a finite set of points, lines or convex sets is forced to contain. The statements are short and need almost no computation, yet the topic supplies some of the hardest IMO problems, because the few methods must survive every degenerate configuration. We assume L4-CO1 and the graph language of L5-CO1. Points are in *general position* if no three of them are collinear.

1.1 Extremal choices

Principle.
Choose an extremal object, such as a closest pair, a triangle of least or greatest area, a point and a line at least positive distance, or a hull side (a line through two points with no points on one side). Then show that a failure of the claim produces a better object.

Theorem 1 (Sylvester-Gallai).
If a finite set S of points does not lie on one line, then some line contains exactly two points of S.

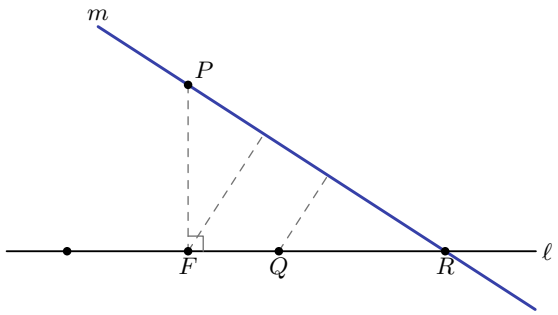
Proof. (Kelly's extremal choice.) Consider pairs (P, ℓ) where ℓ is a line through at least two points of S and P is a point of S off ℓ. Such pairs exist and there are finitely many, so choose one with the distance $d(P, \ell)$ least. Let F be the foot of the perpendicular from P to ℓ, and suppose ℓ contains three points of S. Two of them lie on the same closed ray of ℓ from F, say Q and R with Q on segment FR ($Q = F$ is allowed). The line $m = PR$ does not contain Q. The distance to m falls linearly from F to R, and the altitude from the right angle of triangle PFR is shorter than the leg PF, so

$$d(Q, m) \le d(F, m) = \frac{PF \cdot FR}{PR} < PF = d(P, \ell).$$

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So (Q, m) beats (P, ℓ), a contradiction. ■



A closest pair of points, or a line alone, gets nowhere. The extremal object is a point together with a line, and this mixed choice is worth remembering.

Example 1.
A set S of $n \geq 3$ points does not lie on one line. Prove that at least n different lines pass through two or more points of S.

Solution.
Induct on n. Three points not on a line give three lines. Let $n \geq 4$. By Theorem 1 some line PQ contains no point of S other than P and Q.
If $S \setminus \{P\}$ lies on a line λ, then $P \notin \lambda$, and λ together with the lines PX for $X \in S \setminus \{P\}$ gives n different lines. Otherwise, by induction, $S \setminus \{P\}$ determines at least $n - 1$ lines. None of them is PQ, because PQ contains only one point of $S \setminus \{P\}$. Adding PQ gives at least n. ■
Remark. This is the theorem of de Bruijn and Erdős (1948), and $n - 1$ points on a line with one point off the line show that n is attained. Theorem 1 says which point to remove in the induction.

1.2 Convex position

Points are in *convex position* if each of them is a vertex of their convex hull. Four points in general position fail to be in convex position exactly when one lies inside the triangle of the other three.

Theorem 2 (Klein).
Any five points in general position include four in convex position.

Proof. If the hull has four or five vertices, take four of them. Otherwise the hull is a triangle ABC containing D and E. Line DE meets two sides of the triangle and misses the vertices, so two vertices, say B and C, lie on the same side of DE. If E were inside triangle BCD, the line through the vertex D and the interior point E would separate B from C, and the same holds with D and E exchanged. If B were inside triangle CDE, line BC would separate D from E, but D and E lie inside triangle ABC, on one side of BC. The same holds for C. So B, C, D, E are in convex position. ■

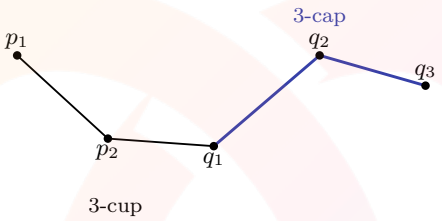
Rotate the axes so that no two points share an x-coordinate. A *k-cup* is a sequence of k points, in increasing order of x, in which the slopes of consecutive segments increase. A *k-cap* is the same with decreasing slopes. Any two points form a 2-cup and a 2-cap, and the points of a cup or a cap are in convex position.

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Theorem 3 (Erdős-Szekeres, 1935).
Let $k, l \geq 2$. Any $\binom{k+l-4}{k-2} + 1$ points in general position with different x-coordinates contain a k-cup or an l-cap. Hence any $\binom{2k-4}{k-2} + 1$ points in general position contain k points in convex position.

Proof. Write $f(k,l) = \binom{k+l-4}{k-2} + 1$ and induct on $k+l$. If $k = 2$ or $l = 2$ then $f = 2$ and the claim is clear. Let $k, l \geq 3$, so $f(k,l) = f(k-1,l) + f(k,l-1) - 1$ by Pascal's rule. Suppose a set X of $f(k,l)$ points has no k-cup and no l-cap. Let A be the set of points of X that are the right end of some $(k-1)$-cup. Then $X \setminus A$ has no $(k-1)$-cup, as the right end of one would lie in A, and no l-cap. By induction $|X \setminus A| \leq f(k-1,l) - 1$, so $|A| \geq f(k,l-1)$. As A has no k-cup, A contains an $(l-1)$-cap $q_1 q_2 \dots q_{l-1}$.

Now $q_1 \in A$ is the right end of a $(k-1)$-cup $p_1 \dots p_{k-2} q_1$. The slopes of $p_{k-2}q_1$ and $q_1 q_2$ differ by general position. If the second is larger, $p_1 \dots p_{k-2} q_1 q_2$ is a k-cup. If the second is smaller, $p_{k-2} q_1 q_2 \dots q_{l-1}$ is an l-cap. Either way we have a contradiction. ■



Erdős and Szekeres conjectured that $2^{k-2} + 1$ points suffice, which is known for $k \leq 6$. For olympiads the gluing step matters more than the constant. Two chains sharing an endpoint combine into a longer chain of one kind or the other.

1.3 Moving lines

A line that turns or slides changes the counts on the two sides of the line one point at a time. A count that changes sign in steps of one must pass through zero, and the whole difficulty is checking that the count never jumps by more.

Theorem 4 (ham sandwich for points).
Let S consist of $2a$ red and $2b$ blue points in general position. Some line through no point of S has a red and b blue points on each side.

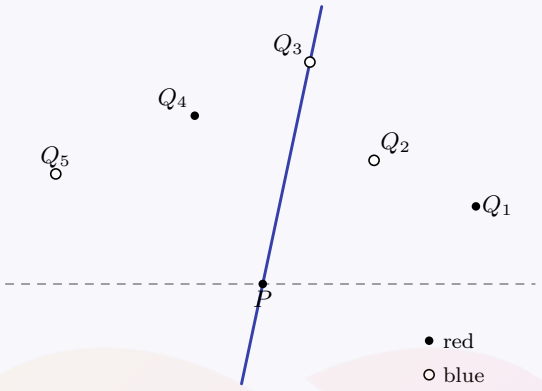
Proof. For an angle θ let $u_\theta = (\cos\theta, \sin\theta)$, and call θ generic if no two points of S have the same projection $X \cdot u_\theta$. All but finitely many θ are generic. For generic θ let L_θ be the $a+b$ points with the smallest projections, separated from the rest by a line perpendicular to u_θ, and let $r(\theta)$ count the red points in L_θ. Since $L_{\theta+\pi} = S \setminus L_\theta$, we have $r(\theta+\pi) = 2a - r(\theta)$.

As θ passes a non-generic value, the order of the projections changes only by swapping pairs of points with equal projections. No three points tie, since they would lie on a line perpendicular to u_θ. At most one of the swapped pairs occupies places $a+b$ and $a+b+1$, so L_θ loses at most one point and gains at most one, and r changes by at most 1. The integer $r(\theta) - a$ is reversed in sign between θ and $\theta+\pi$ and moves in steps of at most 1, so $r(\theta) = a$ for some generic θ. The separating line for that θ works. ■

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The windmill of IMO 2011 Problem 2, on the ladder, pivots a line about the points of S and tracks the same kind of count.

Example 2.
There are n red and n blue points in general position. Prove that some line through a red point and a blue point has, on each side of the line, as many red points as blue points.



Solution.
Let P be a vertex of the convex hull, and say P is red (otherwise swap the colours). Some line through P has every other point on one side, so the other $2n-1$ points, seen from P, lie in an angle less than 180°. List them as $Q_1, \dots, Q_{2n-1}$ in angular order. For each j, the points before Q_j in the list lie on one side of line PQ_j and the points after Q_j lie on the other.
Let s_k be the number of blue points minus the number of red points among $Q_1, \dots, Q_k$. Then $s_0 = 0$, $s_{2n-1} = n - (n-1) = 1$, and consecutive values differ by 1. Let k be least with $s_k = 1$. Then $s_{k-1} = 0$, so Q_k is blue and $Q_1, \dots, Q_{k-1}$ are balanced. The points $Q_{k+1}, \dots, Q_{2n-1}$ contribute $s_{2n-1} - s_k = 0$, so they are balanced too, and line PQ_k works. ■
Remark. The hull vertex turns the rotating line into a walk from 0 to 1. Applying Example 2 again on each side and inducting on n joins the points by n pairwise disjoint red-blue segments.

1.4 Euler's formula and crossings

A *plane graph* is a graph drawn in the plane with edges meeting only at common endpoints. For points in general position joined by segments, two segments without a common endpoint either cross or are disjoint.

Theorem 5 (Euler).
A connected plane graph with v vertices, e edges and f faces, the outer face included, satisfies $v - e + f = 2$. Consequently a plane graph with $v \geq 3$ vertices has at most $3v - 6$ edges.

Proof. Induct on e. If the graph is a tree, then $e = v - 1$ and $f = 1$. Otherwise delete an edge of a cycle. The two faces on either side of that edge merge and the graph stays connected, so e and f both drop by 1. For the bound, join the components by extra edges and assume $e \geq 3$. Every face is then bounded by at least 3 sides of edges, and every edge has two sides, so $3f \leq 2e$. Hence $2 = v - e + f \leq v - \frac{e}{3}$, that is, $e \leq 3v - 6$. ■

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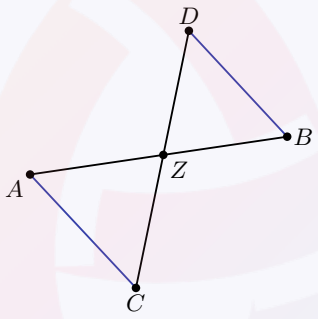
Theorem 6 (crossing lemma).
Let n points in general position be joined by e segments, and let X be the number of crossing pairs of segments. Then $X \geq e - 3n$, and if $e \geq 4n$ then $X \geq \frac{e^3}{64n^2}$.

Proof. Delete one segment from each crossing pair. At least $e - X$ segments remain and form a plane graph, so $e - X \leq 3n$ (for $n \leq 2$ as well). For the second bound keep each point independently with probability $p = \frac{4n}{e} \leq 1$, together with the segments whose two ends are kept. The first bound holds for every outcome, so the same inequality holds for the expected values. A crossing pair has four different ends, so

$$p^4 X \geq p^2 e - 3pn, \qquad \text{so} \qquad X \geq \frac{e}{p^2} - \frac{3n}{p^3} = \frac{e^3}{16n^2} - \frac{3e^3}{64n^2} = \frac{e^3}{64n^2}.$$

■

Example 3.
There are $n \geq 3$ points in the plane, any two at distance at least 1. Prove that at most $3n - 6$ pairs of them are at distance exactly 1.



Solution.
Draw the segment AB for every pair at distance 1. By Theorem 5 we only need to show that two of these segments, AB and CD, meet only at a common endpoint.
Suppose a point Z lies on both segments and is not a common endpoint. If Z is an endpoint of one segment, say $Z = C$, then C is inside segment AB, and $CA + CB = 1$ gives $\min(CA, CB) \leq \frac{1}{2}$, a contradiction. If the two segments lie on one line, they overlap, and two different unit segments on a line that overlap have an endpoint of one inside the other, the case just excluded. In the remaining case Z is inside both segments, the four endpoints are different (a shared endpoint and Z would put both segments on one line), and Z is not on segment AC, since then C would lie on line $AZ = AB$. So $AC < AZ + ZC$, and in the same way $BD < BZ + ZD$. Adding,

$$AC + BD < AB + CD = 2,$$

so AC or BD is less than 1, a contradiction. ■
Remark. Proving that the natural drawing is a plane graph is the usual way into Euler's formula. The bound is not sharp. Harborth (1974) showed that the maximum is $\lfloor 3n - \sqrt{12n - 3} \rfloor$.

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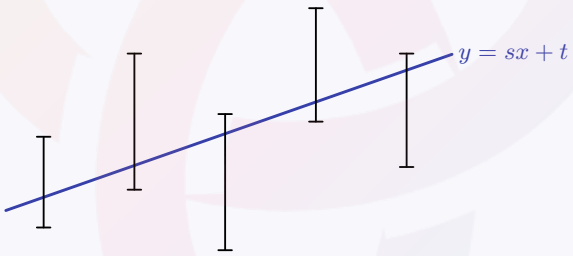
1.5 Helly's theorem

Lemma (Radon).
Any four points in the plane can be split into two groups whose convex hulls meet.
Theorem 7 (Helly). Let $K_1, \dots, K_m$ be convex sets in the plane, $m \geq 3$, such that every three of them have a common point. Then all of them have a common point.

Proof. For the lemma, if one point lies in the triangle, possibly degenerate, of the other three, take that point alone. Otherwise the four points are in convex position and the two diagonals of their quadrilateral cross.

For the theorem induct on m, the case $m = 3$ being the hypothesis. For $m = 4$ choose x_i in every set except K_i, and split $\{x_1, x_2, x_3, x_4\}$ by the lemma into two groups with a common point z of their hulls. For each k, the point x_k is in only one group, and every point of the other group lies in K_k, so $z \in K_k$ by convexity. Hence z lies in all four sets. For $m \geq 5$ replace K_{m-1} and K_m by the convex set $K_{m-1} \cap K_m$. Any three sets of the new family meet, by the hypothesis or by the case $m = 4$, so the induction applies. ■

Example 4.
Finitely many vertical segments lie on different vertical lines, and every three of them are met by a common line. Prove that one line meets all of them.



Solution.
Let segment i be $\{x_i\} \times [c_i, d_i]$. A line meeting two of the segments is not vertical, so we only consider the lines $y = sx + t$. Such a line meets segment i if and only if

$$c_i \leq sx_i + t \leq d_i.$$

Let K_i be the set of points (s, t) satisfying this. Each K_i is a strip between two parallel lines, hence convex, and every three of these sets meet by hypothesis. By Theorem 7 (or directly, for fewer than three segments) all of them have a common point (s, t), and the line $y = sx + t$ meets every segment. ■
Remark. Lines became points and the condition became convex. The real work in a Helly problem is finding the right family.

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1.6 Areas and lattice points

Area pigeonhole.
If sets inside a region of area A have total area greater than kA, some point lies in at least $k+1$ of them.
Theorem 8 (Blichfeldt). A bounded region of area greater than 1 contains two different points whose difference has integer coordinates.
Theorem 9 (Minkowski). A convex region symmetric about the origin O with area greater than 4 contains a lattice point other than O.

Proof. For Theorem 8 cut the region R along the grid lines into the pieces $R \cap (m+Q)$, where m runs over the lattice points and $Q = [0,1)^2$, and translate each piece by $-m$ into Q. The pieces have total area greater than $[Q] = 1$, so two of them, from $m \neq m'$, share a point t. Then $t+m$ and $t+m'$ lie in R. For Theorem 9 apply Theorem 8 to $\frac{1}{2}K$, which has area greater than 1, to get $x \neq y$ in $\frac{1}{2}K$ with $x-y$ a lattice point. Both $2x$ and $-2y$ lie in K, so their midpoint $x-y$ does too. ■

By the shoelace formula every lattice polygon has area in $\frac{1}{2}\mathbb{Z}$. Pick's theorem $[P] = I + \frac{1}{2}B - 1$, for a simple lattice polygon with I lattice points inside and B on the boundary, is proved via Theorem 5 in Aigner and Ziegler, *Proofs from THE BOOK*. .

Common traps.

- Assuming general position when the problem does not give general position.
- In a rotating argument, forgetting that several pairs can swap at the same angle.
- Applying $e \leq 3n-6$ before proving that the drawing has no crossings, including endpoints lying on other segments.
- Applying Helly to sets that are not convex, or to infinitely many sets (the half-planes $x \geq k$ for $k = 1, 2, \ldots$ have no common point).
- Pieces of diameter d give "at most d", not "less than d".

1.7 How to spot this in a contest

1. "Some line, point or triangle of the configuration has a property" suggests an extremal choice. If one kind of object fails, try a pair such as a point and a line.
2. "Many points contain a convex polygon" suggests cups, caps and Theorem 3.
3. "A line or a point divides the set fairly" suggests a rotating line and a count that changes by one.
4. A linear bound on segments or touching pairs suggests a plane graph and Theorem 5. Many crossings suggest Theorem 6.
5. "Every three have a common point" or "every three can be covered" suggests Helly, often in a space of parameters.
6. Lattice points and area suggest the shoelace formula, Pick, or cutting into unit squares and translating.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ A finite set S of points in the plane has the property that any three points of S lie in some strip of width 1, the closed region between two parallel lines at distance 1. Prove that S lies in a strip of width 2, and show that 2 cannot be replaced by 1.

Problem 2 ⋆⋆⋆ A finite set S of at least four points in the plane, no three on a line, has the property that every circle through three points of S passes through a fourth point of S. Prove that all the points of S lie on one circle.

Problem 3 ⋆⋆⋆ Let S be a set of $2n+1$ points in the plane, no three on a line, where $n \geq 1$. A segment joining two points of S is *balanced* if the line through the segment has $n-1$ points of S on one side and n on the other. A line ℓ through no point of S has k points of S on one side, where $k \leq n$. Prove that ℓ crosses exactly $2k$ balanced segments.

Problem 4 ⋆⋆⋆⋆ Prove that among any five lattice points in the plane, no three on a line, there are three that form a triangle of area at least $\frac{3}{2}$. Show that $\frac{3}{2}$ cannot be replaced by a larger number.

Problem 5 ⋆⋆⋆⋆ At least three points in the plane, none of them at the point O, are each coloured red or blue. For any three of the points there is a circle through O with the red points among the three inside or on the circle and the blue points among the three outside the circle. Prove that some circle through O has all the red points inside or on the circle and all the blue points outside. Show that three cannot be replaced by two.

Problem 6 ⋆⋆⋆⋆ Let k be a positive integer and S a set of $n \geq 2k+1$ points in the plane, no three on a line. A point X of S is *k-exposed* if some closed half-plane contains X and at most k points of S, counting X. Prove that at least $2k+1$ points of S are k-exposed, and that for all such k and n exactly $2k+1$ points can be k-exposed.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1994 Problem 7** ⋆⋆
   List the lattice points on $x^2+y^2=50$, then think about which lines through them or tangent to the circle are allowed.

2. **Japan MO 1997 (AoPS wiki, Pigeonhole Principle, Olympiad Problems)** ⋆⋆
   Cut the disc into nine regions, each of diameter less than 2, and count.

3. **Baltic Way 1992 Problem 4** ⋆⋆⋆
   Write each squared side as a sum of two squares of integers and track parities around the hexagon.

4. **USAMTS Year 18 Round 1 Problem 4** ⋆⋆⋆
   Pigeonhole on a suitable finite grid of points, first on one line, then on several parallel lines.

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5. **Baltic Way 2003 Problem 10** ⋆⋆⋆
   Reduce coordinates modulo 4 and ask how many points of a given residue class you can keep.

6. **BMO2 2020/21 Problem 2** ⋆⋆⋆
   *Think about which side lengths n can be written as a*x+b*y in two ways, and what that does to the tiling.*

7. **IMO Longlist 1959-66 Czechoslovakia 1** ⋆⋆⋆
   Start from an edge of the convex hull and choose the third point extremally.

8. **BMO1 2017/18 Problem 5** ⋆⋆⋆
   Look for many disjoint small regions of the 200 x 200 board, each of which must contain a coloured square to destroy every comb inside that region.

9. **IMO 1989 Problem 3** ⋆⋆⋆⋆
   Count pairs (point, pair of points equidistant from that point) in two different ways.

10. **BMO2 2014/15 Problem 4** ⋆⋆⋆⋆
    Try small cases of an n-loop for the first few n, then look for a way to extend a loop by two points.

11. **IMO 1969 Problem 5** ⋆⋆⋆⋆
    First show that among any five points, no three collinear, some four form a convex quadrilateral, then count by double counting.

12. **IMO 2011 Problem 2** ⋆⋆⋆⋆
    Orient the line and start with a line through P that has equally many points on each side, or one more on one side. Show that the count on each side never changes as the windmill turns.

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4 Hints

1. Take a triangle of greatest area with vertices in S. First show that a triangle inside a strip of width 1 has an altitude of length at most 1.

2. Invert about a point of S. What do the circles through the centre of inversion become, and which theorem of the lesson then applies?

3. Slide a line parallel to ℓ in from far away and watch the count change each time the line passes a point P of S. To find that change, turn a line about P through half a turn.

4. Two of the five points have a lattice midpoint. What can you say about the area of a triangle with these two points as vertices?

5. A circle through O is determined by the centre c. For a single point P, which centres c put P inside or on the circle?

6. For a direction u let $F(u)$ be the k points of S with the smallest values of $X \cdot u$. If only $2k$ points were exposed, what would $F(u) \cup F(-u)$ be for every u? Now turn u through half a turn.

5 Solutions

Problem 1

A finite set S of points in the plane has the property that any three points of S lie in some strip of width 1, the closed region between two parallel lines at distance 1. Prove that S lies in a strip of width 2, and show that 2 cannot be replaced by 1.

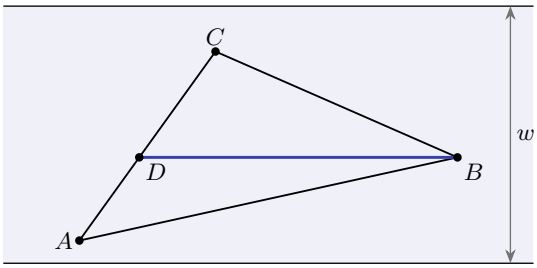
If S lies on a line, a strip of any width around that line works. Otherwise we first prove a lemma.

Lemma. If a triangle ABC lies in a strip of width w, then the altitude to the longest side of the triangle is at most w.

Proof. Relabel so that the distances of A, B, C from one boundary line of the strip are $a \le b \le c$. Then $c - a \le w$. The line through B parallel to the strip meets side AC at a point D. Triangles ABD and CBD share the base BD, with heights $b - a$ and $c - b$, so

$$[ABC] = \tfrac{1}{2}\, BD\,(c - a) \le \tfrac{1}{2}\, BD \cdot w.$$

The closed disc with centre B and radius $\max(BA, BC)$ is convex and contains A and C, so the disc contains D, and BD is at most the longest side L. Since $[ABC] = \tfrac{1}{2}Lh$, where h is the altitude to the longest side, we get $h \le w$. ■



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Now choose a triangle ABC of greatest area with vertices in S. The area is positive because S is not on a line. Let BC be a longest side and h the altitude from A, so $h \le 1$ by the lemma. Every $X \in S$ satisfies $[XBC] \le [ABC]$, so X is at distance at most h from line BC. Hence S lies in the strip of width $2h \le 2$ whose middle line is BC.

For the second part take the four vertices of a square of side $\sqrt{2}$. Any three of them form a right isosceles triangle with legs $\sqrt{2}$ and hypotenuse 2, so the altitude to the hypotenuse is 1, and the three points lie in the strip of width 1 between the hypotenuse and the parallel line through the right-angled vertex. With the sides of the square along the axes, a strip with normal direction $(\cos\varphi, \sin\varphi)$ containing the square has width at least

$$\sqrt{2}\,(|\cos\varphi| + |\sin\varphi|) \ge \sqrt{2} > 1,$$

since $(|\cos\varphi| + |\sin\varphi|)^2 = 1 + |\sin 2\varphi| \ge 1$. ■

So the best constant lies between $\sqrt{2}$ and 2.

Problem 2

A finite set S of at least four points in the plane, no three on a line, has the property that every circle through three points of S passes through a fourth point of S. Prove that all the points of S lie on one circle.

Fix $P \in S$ and invert about P with any radius, writing X' for the image of a point $X \ne P$. Recall that inversion is a bijection of the plane with P removed, that X' lies on ray PX, and that the circles through P correspond to the lines not through P. Let $T = \{X' : X \in S \setminus \{P\}\}$, a set of at least three points.

Take two points $Q' \ne R'$ of T. The points P, Q, R are not collinear, and Q', R' lie on the rays PQ, PR, so line $Q'R'$ does not pass through P. Hence line $Q'R'$ is the image of the circle through P, Q, R. By hypothesis this circle contains a fourth point U of S, and then $U' \in T$ lies on line $Q'R'$ and differs from Q' and R'. So every line through two points of T contains a third point of T, and by Theorem 1 the set T lies on a line λ.

The line λ does not pass through P, since we saw that P is never collinear with two points of T. So λ is the image of a circle ω through P, and every point of $S \setminus \{P\}$ lies on ω. Together with P, the whole of S lies on ω. ■

Remark. Inversion about a point of the set carries Theorem 1 from lines to circles through that point.

Problem 3

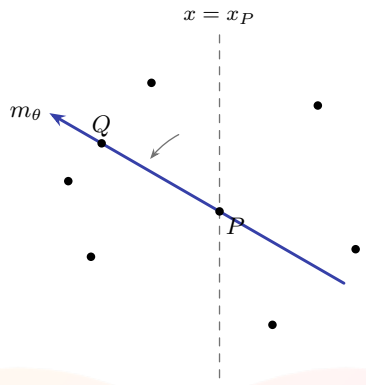
Let S be a set of $2n+1$ points in the plane, no three on a line, where $n \ge 1$. A segment joining two points of S is balanced *if the line through the segment has $n-1$ points of S on one side and n on the other. A line ℓ through no point of S has k points of S on one side, where $k \le n$. Prove that ℓ crosses exactly $2k$ balanced segments.*

Since ℓ passes through no point of S, a segment crosses ℓ if and only if the endpoints lie on opposite sides of ℓ. So the count depends only on how ℓ splits S. Turning ℓ slightly about a point of ℓ keeps this split, as ℓ is at positive distance from every point of S. Hence we may assume that ℓ is parallel to no segment joining two points of S, and choose axes with ℓ vertical. Then no two points of S have the same x-coordinate.

Lemma. Let $P \in S$ have j points of S to the left of P. Let r and l be the numbers of balanced segments joining P to a point right of P and to a point left of P. Then $r - l$ is 2 if $j < n$, 0 if $j = n$, and -2 if $j > n$.

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Proof. For $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$ let m_θ be the line through P oriented by $u_\theta = (\cos\theta, \sin\theta)$. When m_θ contains no other point of S, let $g(\theta)$ be the number of points of S on the left of m_θ minus the number on the right. The $2n$ points other than P each contribute ± 1, so g is even. The line $m_{\pi/2}$ points upwards, and the points on the left of $m_{\pi/2}$ are the j points left of P, so $g(\frac{\pi}{2}) = 2j - 2n$. The line $m_{3\pi/2}$ is the same line reversed, so $g(\frac{3\pi}{2}) = 2n - 2j$.



As θ increases, the line meets each $Q \in S \setminus \{P\}$ once, and never two points at once, since no three points of S are collinear. A point Q left of P is met by the ray $\{P + tu_\theta : t > 0\}$, which turns into the left side of the line. So Q passes from the left side to the right side and g drops by 2. A point right of P is met by the opposite ray, and g rises by 2. Line PQ is balanced if and only if the other $2n - 1$ points contribute ± 1 to g, that is, if and only if one of the values of g just before and just after Q is 0. So l counts the drops of g that start or end at 0, and r counts such rises.

The steps that start or end at 0 are those that cross the level 1 or the level -1. The walk g goes from $a = 2j - 2n$ to $-a$. If $a < 0$, the walk starts below both levels and ends above both, so each level is crossed upwards once more than downwards, and $r - l = 2$. If $a = 0$, the walk starts and ends at 0, so each level is crossed as often upwards as downwards, and $r - l = 0$. The case $a > 0$ gives $r - l = -2$ in the same way. ■

Now slide a vertical line from far left to ℓ, and let N be the number of balanced segments that the moving line crosses, so $N = 0$ at the start. The value of N changes only when the line passes a point $P \in S$. At that moment the balanced segments from P to the left stop crossing the line and those from P to the right start, so N changes by $r - l$. The point passed after i others has $j = i$, so by the lemma $N = 2i$ once $i \le n$ points are passed, $N = 2n$ once $n + 1$ are passed, and N falls by 2 at each later point. Hence $N = 2\min(i, 2n + 1 - i)$ after i points. The line ℓ has k points on one side and $2n + 1 - k > k$ on the other, so ℓ crosses exactly $2k$ balanced segments. ■

Remark. The same sweep for $2n$ points shows that a line with $k \le n$ points on one side crosses exactly k of the segments whose lines have $n - 1$ points on each side.

Problem 4

Prove that among any five lattice points in the plane, no three on a line, there are three that form a triangle of area at least $\frac{3}{2}$. Show that $\frac{3}{2}$ cannot be replaced by a larger number.

Write $(p, q) \times (r, s) = ps - qr$. By the shoelace formula $2[XYZ] = |(Y - X) \times (Z - X)|$, so twice the area of a lattice triangle is an integer, and we need a triangle with $2[XYZ] \ge 3$.

The parities of the two coordinates give four classes, so two of the five points, A and B, lie in the same class, and $B - A = 2v$ for a lattice vector v. For each of the other three points X,

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$$[ABX] = \tfrac{1}{2}\,|2v \times (X - A)| = |v \times (X - A)|\,,$$

a positive integer since A, B, X are not collinear. If one of these areas is at least 2 we are done. Otherwise $v \times (X - A) = \pm 1$ for all three points X. Then the coordinates p, q of v are coprime, as a common factor would divide $v \times (X - A)$. Choose integers s, t with $pt - qs = 1$ and put $w = (s, t)$, so $v \times w = 1$.

Every lattice vector (a, b) equals $xv + yw$ with the integers $x = at - bs$ and $y = pb - qa$, and

$$(xv + yw) \times (x'v + y'w) = xy' - x'y.$$

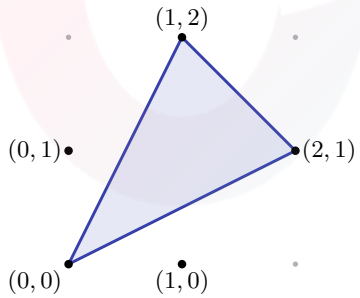
So we may give the point $A + xv + yw$ the new coordinates (x, y) without changing any area. Now $A = (0, 0)$, $B = (2, 0)$, and the other three points have $y = v \times (X - A) = \pm 1$. No three of them lie on one line $y = \pm 1$, so after the reflection $y \mapsto -y$ we may take $C = (a, 1)$, $D = (b, 1)$ and $E = (c, -1)$ with $b > a$.

If $b \geq a + 2$, then $2[CDE] = 2(b - a) \geq 4$. Otherwise $b = a + 1$, and with $m = a + c$ the shoelace formula gives

$$2[ACE] = |m|, \qquad 2[ADE] = |m + 1|, \qquad 2[BDE] = |3 - m|, \qquad 2[BCE] = |4 - m|.$$

If all four were at most 2, then $m + 1 \leq 2$ and $4 - m \leq 2$, so $m \leq 1$ and $m \geq 2$, which is absurd. So some triangle has $2[XYZ] \geq 3$.

For the second part take $(0, 0)$, $(1, 0)$, $(0, 1)$, $(2, 1)$ and $(1, 2)$, no three of which are collinear. Of the ten triangles, three have area $\frac{1}{2}$, six have area 1, and the triangle with vertices $(0, 0)$, $(2, 1)$, $(1, 2)$ has area $\frac{3}{2}$. ■



The five points of the example are the vertices of a convex pentagon of area $\frac{5}{2}$, the least area of a convex lattice pentagon.

Problem 5

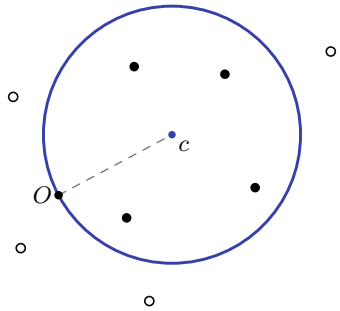
At least three points in the plane, none of them at the point O, are each coloured red or blue. For any three of the points there is a circle through O with the red points among the three inside or on the circle and the blue points among the three outside the circle. Prove that some circle through O has all the red points inside or on the circle and all the blue points outside. Show that three cannot be replaced by two.

Treat points as vectors from O. A circle through O has a centre $c \neq O$ and radius $|c|$, and a point P is inside or on this circle if and only if $|P - c|^2 \leq |c|^2$, that is,

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$$2P \cdot c \geq |P|^2.$$

For a red point P let K_P be the closed half-plane of centres c with $2P \cdot c \geq |P|^2$. For a blue point Q let K_Q be the open half-plane of centres c with $2Q \cdot c < |Q|^2$. Then a centre $c \neq O$ gives a circle that works for some of the points if and only if c lies in all of their sets.



By hypothesis any three of these convex sets have a common point. So do any one or two of them, as we may add further points to make three. By Theorem 7 some point c lies in every set. If some point P is red, then $c \neq O$, because $|P|^2 > 0$ puts O outside K_P, and the circle with centre c through O works. If every point is blue, then O lies in each of the finitely many open sets K_Q, so a small disc about O lies in all of them, and any centre $c \neq O$ in that disc works.

For the last part let three red points be the vertices of an equilateral triangle with centre O. Any two of them lie on the circle through O and these two points. A closed disc containing all three contains the triangle, so O is an interior point of the disc and does not lie on the circle. ■

Remark. Inversion about O turns the circles through O into lines. Red points go to the side of the line away from O and blue points to the side of O, so O behaves as one more blue point.

Problem 6

Let k be a positive integer and S a set of $n \geq 2k+1$ points in the plane, no three on a line. A point X of S is k-exposed *if some closed half-plane contains X and at most k points of S, counting X. Prove that at least $2k+1$ points of S are k-exposed, and that for all such k and n exactly $2k+1$ points can be k-exposed.*

Call a unit vector u generic if no two points of S have the same value of $X \cdot u$. All but finitely many u are generic. For generic u let $F(u)$ be the set of the k points of S with the smallest values of $X \cdot u$. Each $X \in F(u)$ is k-exposed, as the closed half-plane $\{Y : Y \cdot u \leq X \cdot u\}$ contains X and at most k points of S. Since $n > 2k$, the sets $F(u)$ and $F(-u)$ are disjoint.

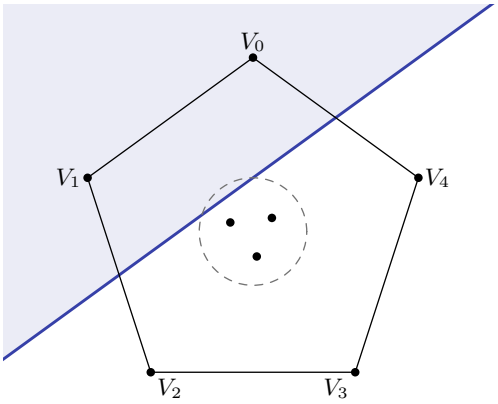
Suppose that the set A of k-exposed points has at most $2k$ elements. For generic u the set $F(u) \cup F(-u)$ has $2k$ elements and lies in A, so

$$F(u) \cup F(-u) = A \quad \text{for every generic } u.$$

Turn u through half a turn from a generic u_0 to $-u_0$. The set $F(u)$ changes on the way, because $F(u_0)$ and $F(-u_0)$ are disjoint. As in the proof of Theorem 4, at a non-generic direction the order of the values $X \cdot u$ changes only by swaps of two points with equal values, because no three points of S lie on a line perpendicular to u. So $F(u)$ changes only when the points in places k and $k+1$ swap. Just after such a swap, the point X that moved to place $k+1$ is not in $F(u)$, and not in $F(-u)$, which consists of places $n-k+1, \ldots, n$, where $n-k+1 > k+1$. But X was in $F(u)$ just before, so $X \in A$, a contradiction. Hence at least $2k+1$ points are k-exposed.

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For the second part let $V_0, \ldots, V_{2k}$ be the vertices of a regular $(2k+1)$-gon with centre O and circumradius 1. Add $n - 2k - 1$ points within distance $\rho = \sin \frac{\pi}{4k+2}$ of O, with no three of all n points on a line. Each V_i is k-exposed, since the closed half-plane beyond the tangent to the circumcircle at V_i contains no other point.



Let Z be an added point and $H = \{Y : Y \cdot u \geq t\}$ a closed half-plane containing Z, with u a unit vector. Then $t \leq Z \cdot u \leq OZ < \rho$. Since $\rho = \cos \frac{k\pi}{2k+1}$, the vertices with $V_i \cdot u \geq \rho$ are the vertices on the closed arc of the circumcircle of angular length $k \cdot \frac{2\pi}{2k+1}$ centred in the direction u. Consecutive vertices are $\frac{2\pi}{2k+1}$ apart, so this arc contains at least k vertices, and all of them lie in H. So H contains Z and at least k further points, and Z is not k-exposed. Exactly $2k+1$ points are k-exposed. ■

For $k = 1$ the result says that the convex hull has at least three vertices. General position matters, as n points on a line have only $2k$ k-exposed points.

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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